Security Log Investigation

Incident Analysis Report — Synthetic Authentication Log

Prepared from authentication-log-sample.csv • Review window: 1 August 2026, 09:00–11:47 UTC • Records reviewed: 12

Executive summary
The sample contains two suspicious authentication patterns: three rapid failures against account stu-117 from 198.51.100.27, and four failures across four accounts from 203.0.113.42 within 12 seconds. The log also records an account-lock event and a rate-limit block. These patterns warrant follow-up, but the supplied records do not prove successful compromise.

1. Scope and method

The 12 CSV records were sorted chronologically. Timestamps were interpreted as UTC because they end in “Z”; text fields were trimmed and normalized to lowercase where appropriate. Detection checks focused on repeated failures to one account, one IP targeting multiple accounts, recorded protective-control events, and later successful authentication for an account seen in earlier failures.

2. Key findings

ID / Priority	Evidence observed	Assessment
DET-01 / Medium	3 failed password_login events in ~34 seconds	Possible brute-force activity; confirm policy thresholds and check adjacent logs.
DET-02 / Medium	4 failed password_login events across 4 distinct accounts in 12 seconds	Possible password spraying or account enumeration; password spraying usually reuses a sm
DET-03 / Medium	rate_limit result=blocked at 09:22:01 UTC	Control blocked an event; inspect rate-limit configuration and surrounding events.
DET-04 / Medium	account_lock result=success at 09:05:03 UTC	Account lock was recorded after repeated failures; verify lock duration and legitimate-user im
DET-05 / Low-Medium	Successful password login at 11:47:09 UTC after a failed login for stu-311 at 09:21:17 UTC from a different IP	Correlation for stu-311 only; compromise, device, MFA, and expected locat

3. Incident timeline

Time (UTC)	Event summary
2026-08-01 09:00:12	stu-104 • password_login=success • 192.0.2.18 • android
2026-08-01 09:03:44	stu-117 • password_login=failure • 198.51.100.27 • windows
2026-08-01 09:04:01	stu-117 • password_login=failure • 198.51.100.27 • windows
2026-08-01 09:04:18	stu-117 • password_login=failure • 198.51.100.27 • windows
2026-08-01 09:05:03	stu-117 • account_lock=success • 198.51.100.27 • server
2026-08-01 09:21:10	stu-209 • password_login=failure • 203.0.113.42 • unknown
2026-08-01 09:21:13	stu-311 • password_login=failure • 203.0.113.42 • unknown
2026-08-01 09:21:17	stu-415 • password_login=failure • 203.0.113.42 • unknown
2026-08-01 09:21:22	stu-502 • password_login=failure • 203.0.113.42 • unknown
2026-08-01 09:22:01	stu-209 • rate_limit=blocked • 203.0.113.42 • server
2026-08-01 10:15:32	stu-104 • refresh_token=success • 192.0.2.18 • android
2026-08-01 11:47:09	stu-311 • password_login=success • 192.0.2.99 • ios

4. Recommended response actions

• Preserve the original CSV and retain a copy of the investigation spreadsheet as evidence.

• Review adjacent authentication logs for the same accounts and IPs over a wider time window.

• Confirm whether stu-117’s lock was expected and whether the account owner experienced login issues; follow organizational account-recovery policy.

• Review rate-limit thresholds and confirm the block was generated as intended.

• For stu-311, verify the later successful login with the account owner and available MFA/device/session records before concluding that access was unauthorized.

• If corroborating evidence supports compromise, follow the organization’s incident-response process: secure the account, revoke sessions/tokens as appropriate, reset credentials through approved procedures, and document actions.

5. Detection rules used

R1 — Repeated failed logins to one account: Group by source_ip + user_id; flag >=3 password_login failures within 2 minutes. *Caveat:* Detects rapid repeated attempts; tune threshold to environment.

R2 — One source IP targets multiple accounts: Group by source_ip; flag >=4 distinct user_id values with password_login failure within 2 minutes. *Caveat:* Possible spraying/enumeration. Password reuse cannot be inferred from these fields.

R3 — Rate limiting: Flag event=rate_limit and result=blocked. *Caveat:* Control signal, not proof of attack success.

R4 — Account lock: Flag event=account_lock and result=success. *Caveat:* Control signal; correlate with preceding failed logins.

R5 — Successful login after failures: Flag success after failures for same account, especially from a different IP/device. *Caveat:* Requires context; success alone is not proof of compromise.

6. Limitations and confidence

The dataset is small and lacks MFA outcomes, user-agent, session identifiers, geo-location, and attempted-password details. The IPs shown are from documentation-only IPv4 ranges (192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24); they must not be attributed to real attackers. “Password spraying” is only a hypothesis because the log does not show password values. Findings are preliminary indicators, not a confirmed incident.

7. Conclusion

The strongest evidence is a burst of failures to one account and a second burst from one IP across four accounts, followed by recorded defensive controls. Further correlation is required to determine whether these events reflect malicious activity, benign user errors, or test behavior.

Educational investigation report based only on the supplied synthetic dataset. Not a claim of a real-world security incident.